Services

Services contain the business logic and integration code that controllers call.

Email Service

File: `backend/src/services/emailService.js`

Responsibilities

- Sending ticket lifecycle emails via SMTP
- Starting and managing IMAP monitoring
- Processing incoming emails and creating ticket replies

Key Functions

Function	Purpose
`sendTicketCreatedEmail(ticket)`	Send confirmation when ticket is created
`sendTicketUpdatedEmail(ticket, changes)`	Notify requester of status/priority changes

Function	Purpose
`sendTicketResolvedEmail(ticket)`	Announce resolution and request CSAT
`startEmailMonitoring()`	Begin IMAP polling loop
`processIncomingEmail(email)`	Parse and route incoming email to ticket

Email Template Resolution

Templates are loaded from the `EmailTemplate` model in the database. If a template is missing, the service falls back to a hardcoded default.

Active Directory Service

File: `backend/src/services/adService.js`

Responsibilities

- Authenticating users against LDAP/Active Directory
- Syncing users from AD to the local database
- Syncing directorates (organizational units/groups) as departments

Key Functions

Function	Purpose
`authenticateLDAP(username, password)`	Validate credentials against AD
`syncUsers()`	Import users from AD
`syncDirectorates()`	Import directorates as departments

LDAP Configuration

LDAP settings are stored in `SystemSettings` and include:

- `ldapUrl` — LDAPS server URL
- `ldapBindDn` — Service account DN
- `ldapBindPassword` — Service account password
- `ldapUserSearchBase` — Base DN for user searches
- `ldapUserFilter` — LDAP filter for user objects
- `ldapUsernameAttribute` — Attribute mapped to username
- `ldapEmailAttribute` — Attribute mapped to email
- `ldapNameAttribute` — Attribute mapped to display name

Scheduler

File: `backend/src/services/scheduler.js`

Responsibilities

- Running periodic background jobs using `node-cron`

Scheduled Jobs

Job	Schedule	Purpose
SLA Check	Every 5 minutes	Identify overdue tickets and create notifications
Daily Digest	Daily at 8:00 AM	Send daily ticket summary to staff
AD Sync	Daily at 2:00 AM	Sync users and directorates from AD (if enabled)

Job Lifecycle

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